

Z-transform

⇒ Z-transform - The z-transform of a discrete time signal $x(n)$ is represented by $X(z)$.

$$x(n) \xleftrightarrow{ZT} X(z)$$

Where,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

⇒ Inverse Z-transform -

We know that the relation b/w Z.T. and D.F.T. is,

$$X(z) = \text{DFT} [x(n) \cdot \delta^n]$$

$$\text{IDFT} [X(z)] = x(n) \cdot \delta^n$$

$$x(n) = \delta^n [\text{IDFT} (X(z))]$$

$$= \delta^n \left[\frac{1}{2\pi} \int X(z) e^{j\omega n} d\omega \right]$$

$$= \frac{1}{2\pi} \int X(z) \cdot (\delta e^{j\omega})^n d\omega$$

$$= \frac{1}{2\pi} \int X(z) \cdot z^n \cdot \frac{dz}{jz}$$

$$x(n) = \frac{1}{2\pi j} \int X(z) \cdot z^{n-1} dz$$

$$\left\{ \because X(z) = \text{DFT} (x(n) \delta^n) \right\}$$

$$\left\{ \because \text{IDFT} (X(z)) = \frac{1}{2\pi} \int X(z) \cdot e^{j\omega n} d\omega \right\}$$

$$\left\{ \begin{array}{l} \because z = \delta e^{j\omega} \\ dz = \delta e^{j\omega} \cdot j d\omega \\ dz = jz d\omega \\ \frac{d\omega}{e} = \frac{dz}{jz} \end{array} \right.$$

⇒ Relation b/w DFT & Z.T.

$$\because \text{Z-transform, } X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \quad [\because z = \delta e^{j\omega}]$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot (\delta e^{j\omega})^{-n} = \sum_{n=-\infty}^{\infty} x(n) \cdot \delta^{-n} \cdot e^{-j\omega n}$$

$$\left\{ \because \sum_{n=-\infty}^{\infty} \delta^{-n} e^{-j\omega n} = \text{discrete} \right\}$$

$$X(z) = \text{DFT} [x(n) \cdot \delta^n]$$

⇒ Z-transform of unit step function.

$$\because x(n) = u(n)$$

$$X(z) = \text{Z.T.} [x(n)] = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} u(n) \cdot z^{-n}$$

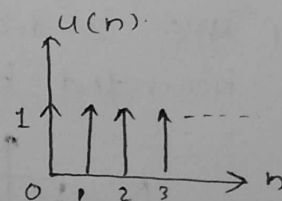
$$\because u(n) = 1$$

$$X(z) = \sum_{n=-\infty}^{\infty} z^{-n}$$

$$\because \sum_{n=0}^{\infty} a^{-n} = \frac{1}{1-a}$$

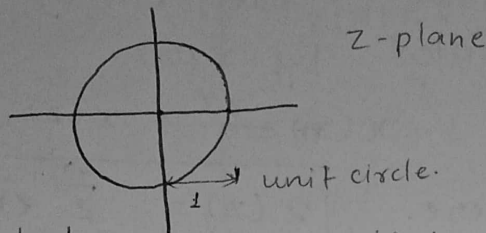
$$\therefore X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$$

$$X(z) = \frac{z}{z-1}$$

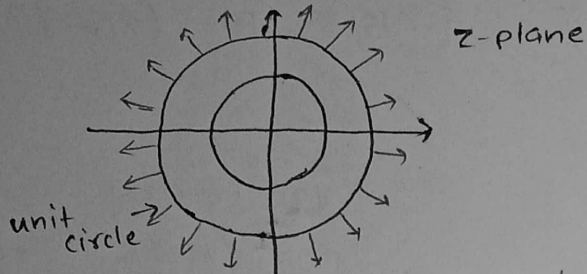


⇒ Region of convergence for z-transform -

The ROC in z-transform is indicated as circles contain unit circle.



→ if $x(n)$ is a right sided sequence & if $|z| = r$ in the ROC, Then all finite value of z , for which $|z| > r$ will also be in ROC.

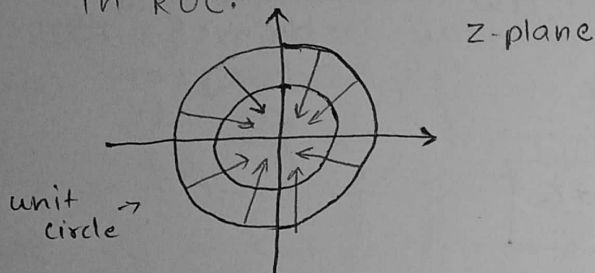


$$x(n) = f(n) \cdot u(n)$$

$$x(n) \longleftrightarrow x(z) = \frac{z}{z-r}$$

$$\text{ROC: } |z| > r$$

→ if $x(n)$ is left sided sequence & if $|z| = r$ in the ROC, then all finite value of z , for which $|z| < r$ will also be in ROC.

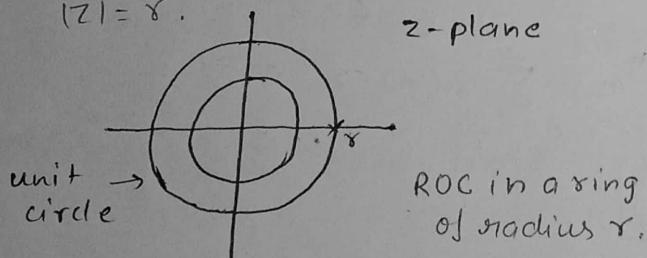


$$x(n) = f(n) \cdot u(-n)$$

$$x(n) \longleftrightarrow x(z) = \frac{z}{z-r}$$

$$\text{ROC: } |z| < r$$

→ if $x(n)$ is two sided signal & if $|z| = r$ circle is in ROC, Then the ROC will contain a ring in z-plane that include $|z| = r$.

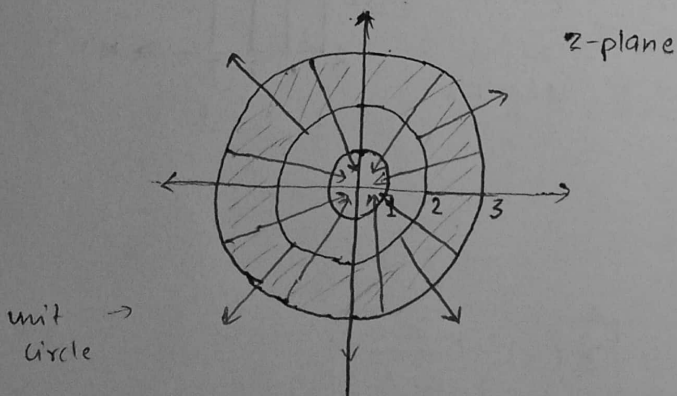


$$x(n) = f_1 u(n) + f_2 u(-n)$$

$$x(n) \longleftrightarrow x(z) = \frac{z}{z-r}$$

$$\text{ROC: } |z| = r$$

→ if the z-transform of $x(n)$ is rational. Then, its ROC is bounded by poles or extended to ∞ .



$$x(n) = 2^n u(n) - 3^n u(n-1)$$

$$\frac{z}{z-2}$$

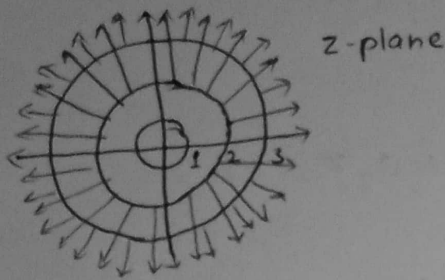
$$|z| > 2$$

$$\frac{z}{z-3}$$

$$|z| < 3$$

$$\text{ROC: } 2 < |z| < 3$$

→ if the Z-transform of $x(n)$ is rational & right sided then ROC is z-plane outside the outermost pole.

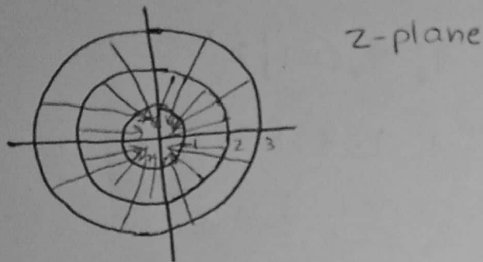


$$x(n) = 2^n u(n) + 3^n u(n)$$

$$\frac{z}{z-2} \quad \frac{z}{z-3}$$

$$\text{ROC: } |z| > 3.$$

→ if the Z-transform of $x(n)$ is rational & left sided, then ROC is the region in z-plane inside the innermost pole.



$$x(n) = -2^n u(-n-1)$$

$$\frac{z}{z-2} \quad \frac{z}{z-3}$$

$$|z| < 2 \quad |z| < 3$$

$$\text{ROC: } |z| < 2$$

⇒ Properties of ROC -

• ROC of impulse function in z-transform -

$$1) \quad \delta(n) \xrightarrow{\text{Z.T.}} 1$$

ROC: Entire z-plane, including $z=0$ & $z=\infty$

$$2) \quad \delta(n-k) \xrightarrow{\text{Z.T.}} z^{-k}$$

ROC: Entire z-plane, including $z=\infty$ & excluding $z=0$

$$3) \quad \delta(n+k) \xrightarrow{\text{Z.T.}} z^k$$

ROC: Entire z-plane, including $z=0$ & excluding $z=\infty$

Eg- ROC of $\delta(n-3) \Rightarrow$ Entire z-plane.

includes $z=\infty$,
exclude $z=0$.

• ROC of discrete time sequence in z-transform.

1) if sequence is purely right sided or causal, then

ROC: Entire z-plane except $z=0$

$$\text{Ex- } x(n) = \{1, 2, 3, 4\} \xrightarrow{\text{Z.T.}} x(z) = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3}$$

2) if sequence is purely left sided or anticausal, then.

ROC: Entire z-plane except at $z=\infty$

$$\text{Ex- } x(n) = \{1, 2, 3, 4\} \xrightarrow{\text{Z.T.}} x(z) = z^3 + 2z^2 + 3z + 4$$

3) if sequence is a two sided, then

ROC: Entire z-plane except at $z=0$ & $z=\infty$

$$\text{Ex } x(n) = \{1, 2, 1, 3, 1\} \xrightarrow{\text{Z.T.}} x(z) = z^2 + 2z + 1 + 3z^{-1} + z^{-2}$$

⇒ Properties of Z-transform-

① linearity - if $x_1(n) \xrightarrow{ZT} x_1(z)$
 $x_2(n) \xrightarrow{ZT} x_2(z)$.

Then linearity -

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow{ZT} a_1 x_1(z) + a_2 x_2(z)$$

Proof.

$$\therefore x(n) = a_1 x_1(n) + a_2 x_2(n)$$

$$\begin{aligned} \therefore X(z) &= Z.T. [x(n)] = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \\ &= \sum_{n=-\infty}^{\infty} [a_1 x_1(n) + a_2 x_2(n)] z^{-n} \\ &= \sum_{n=-\infty}^{\infty} [a_1 x_1(n) z^{-n} + a_2 x_2(n) z^{-n}] \\ &= a_1 \sum_{n=-\infty}^{\infty} x_1(n) z^{-n} + a_2 \sum_{n=-\infty}^{\infty} x_2(n) z^{-n} \end{aligned}$$

$$\boxed{X(z) = a_1 X_1(z) + a_2 X_2(z)}$$

② Time shifting - if $x(n) \xrightarrow{ZT} X(z)$
so time shifting -

$$x(n-m) \xrightarrow{ZT} z^{-m} X(z)$$

$$x(n+m) \xrightarrow{ZT} z^m X(z)$$

Proof.

$$\therefore X(z) = Z.T. [x(n)] = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$\therefore Z.T. [x(n-m)] = \sum_{n=-\infty}^{\infty} x(n-m) \cdot z^{-n}$$

$$\left\{ \text{if } n-m = p \Rightarrow n = p+m \right\}$$

$$Z.T. [x(n-m)] = \sum_{p=-\infty}^{\infty} x(p) z^{-(p+m)}$$

$$= \sum_{p=-\infty}^{\infty} x(p) \cdot z^{-p} z^{-m}$$

$$= z^{-m} \sum_{p=-\infty}^{\infty} x(p) \cdot z^{-p}$$

$$= z^{-m} X(z)$$

$$\boxed{x(n-m) = z^{-m} X(z)}$$

Similarly.

$$\boxed{x(n+m) = z^m X(z)}$$

③ Time reversal - if $x(n) \xrightarrow{Z.T.} X(z)$.

Then time reversal,

$$x(-n) \xrightarrow{Z.T.} X(z^{-1})$$

Proof

$$\therefore X(z) = Z.T. [x(n)] = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$Z.T. [x(-n)] = \sum_{n=-\infty}^{\infty} x(-n) \cdot z^{-n}$$

{if $p = -n$ }

$$Z.T. [x(-n)] = \sum_{n=-\infty}^{\infty} x(p) \cdot z^p = \sum_{p=-\infty}^{\infty} x(p) (z^{-1})^{-p}$$

$$\boxed{Z.T. [x(-n)] = X(z^{-1})}$$

$$x(-n) \xrightarrow{Z.T.} X(z^{-1})$$

④ Time Scaling (Scaling) - if $x(n) \xrightarrow{Z.T.} X(z)$.

Then scaling -

$$a^n x(n) \xrightarrow{Z.T.} X\left(\frac{z}{a}\right)$$

Proof

$$\therefore X(z) = Z.T. [x(n)] = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$\therefore Z.T. [a^n x(n)] = \sum_{n=-\infty}^{\infty} a^n x(n) \cdot z^{-n} = \sum_{n=-\infty}^{\infty} x(n) \cdot (a z^{-1})^n$$

$$= \sum_{n=-\infty}^{\infty} x(n) \cdot \left(\frac{1}{a z^{-1}}\right)^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) \cdot (a^{-1} z)^{-n}$$

$$\boxed{Z.T. [a^n x(n)] = X\left(\frac{z}{a}\right)}$$

⑤ Multiplication & convolution property

$$\text{if } x_1(n) \xrightarrow{Z.T.} X_1(z)$$

$$x_2(n) \xrightarrow{Z.T.} X_2(z)$$

then, multiplication property states that -

$$x_1(n) \cdot x_2(n) \xrightarrow{Z.T.} X_1(z) \cdot X_2(z)$$

and convolution property states that -

$$x_1(n) * x_2(n) \xrightarrow{Z.T.} X_1(z) \cdot X_2(z)$$

⑥ Differentiation property.

$$\text{if } x(n) \xleftrightarrow{z.T.} X(z).$$

Then differentiation property states that

$$n x(n) \xleftrightarrow{z.T.} -z \frac{dX(z)}{dz}$$

⇒ Initial and final value theorem-

initial value theorem- let x be a sequence with z -transform X .

$$x(0) = \lim_{z \rightarrow \infty} X(z).$$

final value theorem- let x be a sequence with z -transform X .

if x is casual and $\lim_{n \rightarrow \infty} x(n)$ exists, then

$$\lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} [(z-1)X(z)]$$

⇒ Power series.

Q $X(z) = \frac{z^3 + 2z^2 + 4z + 8}{z^2}$ for $|z| > 0$.

find inverse z -transform x of the function.

Soln

we can write - $X(z) = z + 2 + 4z^{-1} + 8z^{-2}$.

from the definition of the z -transform,

$$x(n) = \begin{cases} 1 & n = -1 \\ 2 & n = 0 \\ 4 & n = 1 \\ 8 & n = 2 \\ 0 & \text{otherwise.} \end{cases}$$

we have that

$$x(n) = \delta(n+1) + 2\delta(n) + 4\delta(n-1) + 8\delta(n-2)$$

⇒ Characterizing LTI system Using z -transform.

Consider a LTI system with input x , output y , and impulse response h , such a system is characterized by the eqⁿ - $y(n) = x * h(n)$.

Let X , Y and H denote the z -transform of x , y and h , taking the z -transform of both sides of the above eqⁿ and using the time-domain convolution property of z -transform, we have.

$$Y(z) = H(z)X(z)$$

H = transfer function

⇒ Interconnection of LTI System

Here we can drive series and parallel interconnected system.

Let h_1 and h_2 denote the impulse responses of H_1 and H_2 , respectively. The impulse response h of the overall system is given by,

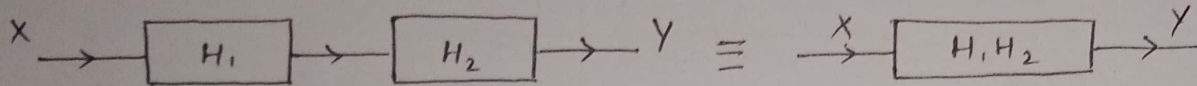
$$h(n) = h_1(n) * h_2(n)$$

taking the z-transform of both sides -

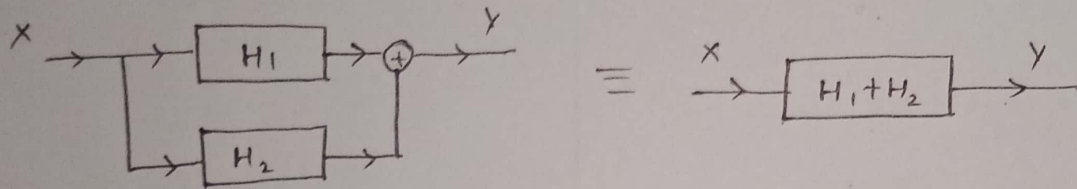
$$H(z) = Z\{h_1 * h_2\}(z)$$

$$= Z h_1(z) Z h_2(z)$$

$$H(z) = H_1(z) H_2(z) \quad (\text{series interconnection})$$



Equivalence involving system funⁿ and the series interconnection of LTI System.



parallel interconnection of LTI System.

The impulse response h of the overall system -

$$h(n) = h_1(n) + h_2(n).$$

Taking z-transform of both sides.

$$H(z) = Z\{h_1 + h_2\}(z)$$

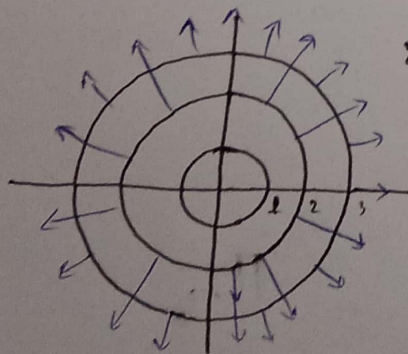
$$= Z h_1(z) + Z h_2(z)$$

$$H(z) = H_1(z) + H_2(z) \quad (\text{parallel connection})$$

⇒ Casuality and Stability check in ROC.

* for casuality -

- (i) The ROC is the exterior of the outside most pole.
- (ii) If $H(z)$ is expressed as ratio of polynomial in 'z'. The order of Numerators can't be greater than order of the denominator.



z-plane

$$|z| > 2$$

$$|z| > 3$$

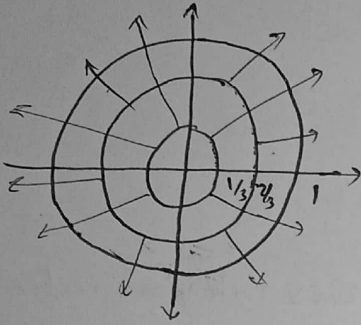
$$\text{ROC: } |z| > 3$$

$$H(z) = \frac{z^2 + z + 1}{z^3 + z + 1}$$

Casual system

* for Stability -

- (i) System function $H(z)$ includes unit circle.
- (ii) A causal LTI system is stable if and only if all the poles of $H(z)$ lies inside the unit circle.



$$|z| > \frac{1}{3}$$

$$|z| > \frac{2}{3}$$

$$\text{ROC: } |z| > \frac{2}{3}$$

⇒ Problems based on z-transform -

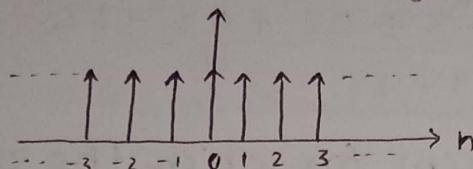
Q. find the z-transform of the signal $a^n u(n)$ & $a^{-n} u(n)$.

Soln

$$\begin{aligned} x(n) &= a^n u(n) \\ \rightarrow x(z) &= z \cdot T. [x(n)] \\ &= \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \\ &= \sum_{n=-\infty}^{\infty} a^n u(n) \cdot z^{-n} \end{aligned}$$

$$\begin{aligned} y(n) &= a^{-n} u(n) \\ y(z) &= z \cdot T. [y(n)] \\ &= \sum_{n=-\infty}^{\infty} y(n) \cdot z^{-n} \\ &= \sum_{n=-\infty}^{\infty} a^{-n} u(n) \cdot z^{-n} \end{aligned}$$

∵ $u(n)$ = unit step signal



$$\begin{aligned} x(z) &= \sum_{n=0}^{\infty} (a^n) (z^{-n}) \\ &= \sum_{n=0}^{\infty} (az^{-1})^n \end{aligned}$$

we know, $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$

$$x(z) = \frac{1}{1-az^{-1}} = \frac{1}{1-(\frac{a}{z})}$$

$$\boxed{x(z) = \frac{z}{z-a}}$$

$$\begin{aligned} y(z) &= \sum_{n=0}^{\infty} a^{-n} z^{-n} \\ &= \sum_{n=0}^{\infty} (az)^{-n} \end{aligned}$$

∵ we know, $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$

$$y(z) = \frac{1}{1-(az)^{-1}} = \frac{az}{az-1}$$

$$\boxed{y(z) = \frac{az}{az-1}}$$

Q. Find the z-transform x of the sequence, $x(n) = -a^n u(-n-1)$.

Soln

$$\therefore x(n) = -a^n u(-n-1)$$

$$\begin{aligned} x(z) &= z \cdot T. [x(n)] = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \\ &= \sum_{n=-\infty}^{\infty} (-a^n u(-n-1)) z^{-n} \end{aligned}$$

$$= - \sum_{n=-\infty}^{\infty} a^n u(-n-1) z^{-n}$$

$$= - \sum_{n=-\infty}^{-1} a^n z^{-n}$$

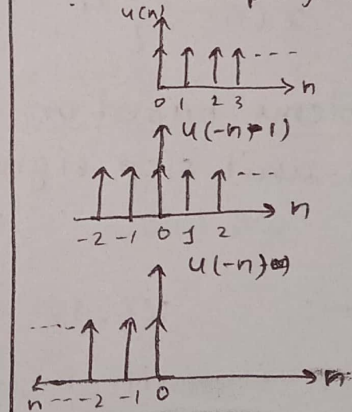
$$= - \sum_{n=1}^{\infty} a^{-n} z^n \quad \left\{ \text{reversal of limits by } n \text{ to } -n \right\}$$

$$= - \sum_{n=1}^{\infty} (a^{-1}z)^n$$

$$x(z) = - \frac{1}{1-a^{-1}z} = - \frac{a^{-1}z}{1-a^{-1}z} \quad \left\{ \because \sum_{n=1}^{\infty} a^n = \frac{1}{1-a} \right\}$$

$$\boxed{x(z) = \frac{z}{z-a}} \quad \text{for } |z| < |a|$$

∵ unit step signal.



Q A finite sequence $x(n)$ is defined as $x(n) = \{5, 3, -3, 0, 4, -2\}$,
find $x(z)$ of given sequence.

Soln

$$\therefore x(n) = \left\{ \overset{x(-2)}{5}, \overset{x(-1)}{3}, \overset{x(0)}{-3}, \overset{x(1)}{0}, \overset{x(2)}{4}, \overset{x(3)}{-2} \right\}$$

z-transform

$$x(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} = \sum_{n=-2}^3 x(n) \cdot z^{-n}$$

$$= x(-2)z^2 + x(-1)z + x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3}$$

$$= 5z^2 + 3z - 3 + 0z^{-1} + 4z^{-2} - 2z^{-3}$$

$$\boxed{x(z) = 5z^2 + 3z - 3 + 4z^{-2} - 2z^{-3}}$$

Q Find the signal $x(n)$ for which the z-transform is
 $x(z) = 4z^4 - z^3 - 3z + 4z^{-1} + 3z^{-2}$

Soln

$$\therefore x(z) = 4z^4 - z^3 - 3z + 4z^{-1} + 3z^{-2}$$

we can written as -

$$x(z) = \underset{x(-4)}{4z^4} - \underset{x(-3)}{z^3} - \underset{x(-2)}{0 \cdot z^2} - \underset{x(-1)}{3z} + \underset{x(0)}{0z^0} + \underset{x(1)}{4z^{-1}} + \underset{x(2)}{3z^{-2}}$$

from z-transform eqn -

$$x(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$x(z) = \sum_{n=-4}^2 x(n) \cdot z^{-n}$$

$$x(n) = \{4, -1, 0, -3, 0, 4, 3\}$$

⇒ Problems based on initial & final value theorem.

Q for unit step signal calculate $x(0)$ & $x(\infty)$.

Soln we know, $u(n) \xleftrightarrow{Z.T.} \frac{z}{z-1}$

$$x(z) = \frac{z}{z-1}$$

then ~~the~~ initial value, $x(0) = \lim_{z \rightarrow \infty} x(z) = \lim_{z \rightarrow \infty} \frac{z}{z-1}$

$$= \lim_{z \rightarrow \infty} \frac{1}{1 - \frac{1}{z}} = \frac{1}{1-0} = 1$$

$$\boxed{x(0) = 1}$$

for final value, $x(\infty) = \lim_{z \rightarrow 1} (z-1)x(z)$

$$= \lim_{z \rightarrow 1} (z-1) \left(\frac{z}{z-1} \right) = \lim_{z \rightarrow 1} z$$

$$\boxed{x(\infty) = 1}$$

Q The z-transform of a anticausal system is

$$x(z) = \frac{3-4z}{1-2z+5z^2}, \text{ find the value of } x(0) \text{ or initial value.}$$

Solⁿ

initial value, $x(0) = \lim_{z \rightarrow \infty} x(z)$

$$= \lim_{z \rightarrow \infty} \left[\frac{3-4z}{1-2z+5z^2} \right] = \lim_{z \rightarrow \infty} \frac{z}{z^2} \left[\frac{\frac{3}{z}-4}{\frac{1}{z^2}-\frac{2}{z}+5} \right]$$

$$x(0) = \lim_{z \rightarrow \infty} \frac{1}{z} \left[\frac{\frac{3}{z}-4}{\frac{1}{z^2}-\frac{2}{z}+5} \right] = 0$$

$$\boxed{x(0)=0}$$

$\Rightarrow$ Problems based on Differentiation property -

Q

$x(z) = \log(1+az^{-1})$, $|z| > |a|$, find $x(n)$?

Solⁿ

$$\Rightarrow \frac{dx(z)}{dz} = \frac{d}{dz} [\log(1+az^{-1})] = \frac{1}{1+az^{-1}} [az^{-1}(-1)] = \frac{-az^{-2}}{1+az^{-1}}$$

$$\Rightarrow -z \frac{dx(z)}{dz} = \frac{az^{-1}}{1+az^{-1}} = \frac{a}{z+a} \xrightarrow{\text{I.Z.T.}} nx(n).$$

$$\Rightarrow a^n u(n) \xrightarrow{\text{Z.T.}} \frac{z}{z-a}$$

$$\Rightarrow (-a)^n u(n) \xrightarrow{\text{Z.T.}} \frac{z}{z+a}$$

$$\Rightarrow (-a)^{n-1} u(n) \xrightarrow{\text{Z.T.}} z^{-1} \left(\frac{z}{z+a} \right) = \frac{1}{z+a}$$

$$\Rightarrow (-a)^n u(n-1) \xrightarrow{\text{Z.T.}} \frac{a}{z+a}$$

$$\Rightarrow nx(n) = (-a)^n u(n-1)$$

$$\boxed{x(n) = \frac{(-a)^n}{n} u(n-1)}$$

$\Rightarrow$ Problems based on multiplication & convolution -

Q

$u(n-1) * \delta(n)$, find z-transform.

Solⁿ

$$\therefore x_1(n) = u(n-1) \xrightarrow{\text{Z.T.}} x_1(z) = \frac{1}{z-1}$$

$$x_2(n) = \delta(n) \xrightarrow{\text{Z.T.}} x_2(z) = 1$$

$$x(z) = \text{Z.T.} [x_1(n) * x_2(n)] = x_1(z) \cdot x_2(z)$$

$$\boxed{x(z) = \frac{1}{(z-1)}}$$

Q

$u(n-2) * \delta(n-3)$, find z-transform.

Solⁿ

$$x_1(n) = u(n-2) \xrightarrow{\text{Z.T.}} x_1(z) = z^{-2} \left(\frac{z}{z-1} \right) = \frac{1}{z} \left(\frac{1}{z-1} \right)$$

$$x_2(n) = \delta(n-3) \xrightarrow{\text{Z.T.}} x_2(z) = z^{-3} \times 1 = \frac{1}{z^3}$$

$$x(z) = \text{Z.T.} [x_1(n) * x_2(n)]$$

$$= x_1(z) \cdot x_2(z)$$

$$x(z) = \frac{1}{z} \left(\frac{1}{z-1} \right) \cdot \frac{1}{z^3} \Rightarrow$$

$$\boxed{x(z) = \frac{1}{z^4(z-1)}}$$

Jitin@BPIT



> Problem based on transfer function & Impulse response -

2. Determine the transfer function and unit impulse response of system using z-transform describe by eqⁿ $\Rightarrow y(n) = \frac{1}{2} y(n-1) + 2x(n)$.

Solⁿ $\therefore$ transfer funⁿ $H(z) = \frac{Y(z)}{X(z)}$

Impulse response $h(n) = \mathcal{Z}^{-1} [H(z)]$

$\therefore y(n) = \frac{1}{2} y(n-1) + 2x(n)$

Taking z-transform of above eqⁿ -

$$Y(z) = \frac{1}{2} z^{-1} Y(z) + 2X(z)$$

$$Y(z) \left[1 - \frac{1}{2} z^{-1} \right] = 2X(z)$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{2}{1 - \frac{1}{2} z^{-1}}$$

$$H(z) = \frac{2z}{z - \frac{1}{2}} = 2 \left[\frac{z}{z - \frac{1}{2}} \right]$$

$$\therefore a^n u(n) \xrightarrow{\text{Z.T.}} \frac{z}{z-a}$$

$$\left(\frac{1}{2}\right)^n u(n) \xrightarrow{\text{Z.T.}} \frac{z}{z - \frac{1}{2}}$$

Taking inverse z-transform -

$$H(n) = 2 \left(\frac{1}{2}\right)^n u(n)$$

$$\boxed{H(n) = (2)^{(1-n)} u(n)}$$

$\Rightarrow$ Problem based on ROC.

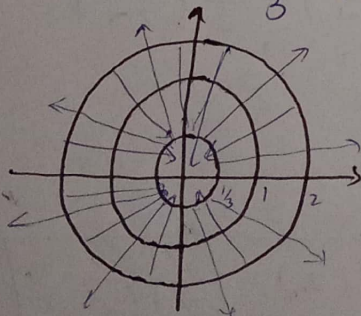
Q. check for stability & causality of a system with response $\left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1)$

Solⁿ $\therefore$ System response $h(n) = \left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1)$

$$H(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z - 2}$$

ROC: $|z| > \frac{1}{3}, |z| < 2$

Common ROC: $\frac{1}{3} < |z| < 2$



Right sided response $\rightarrow$ Left sided response.

$\rightarrow$ ROC includes unit circle so system is stable.

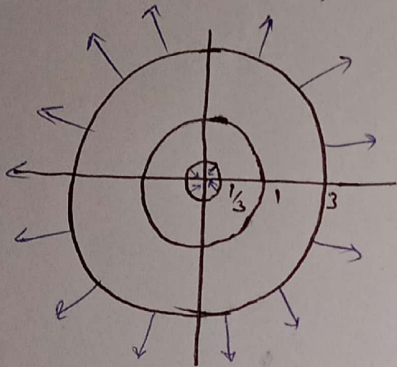
$\rightarrow$ Common ROC is not exterior to centre most pole. so system is anticausal system.

Q find the ROC: $x(n) = 3^{|n|}$

Solⁿ $\therefore x(n) = 3^{|n|} = 3^n u(n) + 3^{-n} u(-n-1)$
 $= 3^n u(n) + \left(\frac{1}{3}\right)^n u(-n-1)$

$$X(z) = \frac{z}{z-3} - \frac{z}{z-1/3}$$

$$\therefore |z| > 3, \quad |z| < \frac{1}{3}$$



→ Here there is no ROC, so we not justify the stability and causality.

for more problems take Akash.